

Uncertainty Analysis of Strain Gauge Measurements Using Digital Data Acquisition

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Abstract— This experiment examined the use of strain-based methods to determine the outside radius of a stool footrest and monitor tiny electrical impulses using digital data acquisition devices. Both unamplified and amplified Wheatstone Bridge voltage signals were used to quantify strain, and the findings were contrasted with a direct geometric radius estimate that was acquired via measuring circumference. The recorded voltage data was statistically analyzed, and the effects of sampling window selection on voltage resolution and signal saturation were investigated. To evaluate the impact of measurement variability, component tolerances, and instrument resolution, an uncertainty analysis was performed using Root Sum Squared propagation. The agreement between measurement methods was compared using expanded uncertainties at the 95% confidence level. The results demonstrated that signal amplification reduced uncertainty in strain-based measurements, while the circumference method provided the most precise radius estimate.

Index Terms— Data acquisition, strain gauge, uncertainty analysis, Wheatstone bridge.

I. INTRODUCTION

THIS experiment explores how digital data acquisition (DAQ) systems can be used to measure small electrical signals and how limitations inherent to those systems affect the reliability of experimental results. While digital acquisition makes it possible to gather data fast and reliably, system features like component tolerances, sampling window selection, signal amplification, and analog-to-digital converter (ADC) resolution have a significant impact on measurement accuracy. This experiment's DAQ divides the entire input range into 16384 discrete levels by digitizing the input voltage using a 14-bit ADC [2]. Consequently, this quantization limits the smallest resolvable change in voltage, and poor input range selection can make it impossible to discern subtle signal fluctuations. Measured values may seem exact but not accurately reflect the system's underlying physical behavior if these aspects are not carefully considered. General DAQ considerations such as input range selection, quantization, and noise floor effects are discussed in more detail in standard DAQ fundamentals references [5].

By attaching a strain gauge to a flexible feeler gauge and integrating the gauge into a Wheatstone Bridge circuit, strain readings were acquired indirectly in this experiment. Bending

the feeler gauge around the stool footrest's outer circle causes surface strain that is proportionate to the footrest's local curvature. Because of this strain, the bridge's electrical resistance changes, which causes a discernible shift in the output voltage. Amplification and careful DAQ sample window selection is necessary to resolve significant strain variations without overloading the measuring device because the unamplified bridge voltage is tiny in comparison to the available ADC range. In addition to the strain-based methods, the footrest circumference was measured using a flexible tape, providing an independent geometric method for estimating the radius and serving as a basis for comparison between techniques.

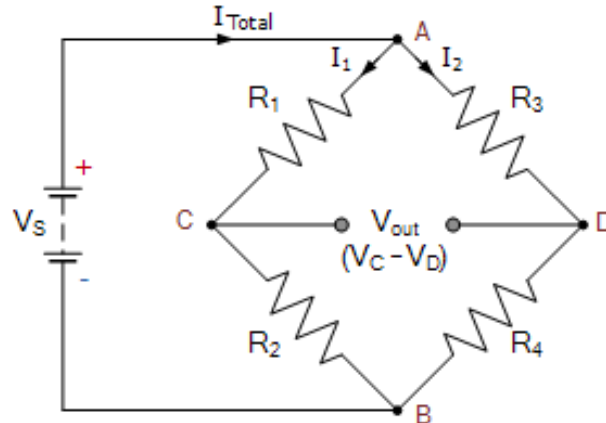


Fig. 1 Wheatstone Bridge schematic showing the configuration used to measure strain in the feeler gauge [6].

Using three separate approaches, strain measurements from the unamplified Wheatstone Bridge voltage, strain measurements from an amplified voltage signal, and a direct circumference measurement, the experiment's goal was to determine the stool footrest's outer radius. Additionally, the effects of sampling window selection on signal saturation and voltage resolution were examined. For each approach, an uncertainty analysis was conducted to take repeatability, signal conditioning, and instrumentation resolution into consideration. To evaluate the dependability of each measurement approach and to determine the degree of agreement between the resulting

radius estimates, expanded uncertainties were presented at the 95% confidence level, as is customary in engineering measurements [3]. Where ε is the measured strain in the strain gauge (mm/mm), ΔV_g is the tared bridge output voltage (V), G_f is the strain gauge factor (unitless), and V_s is the bridge supply voltage (V) (1).

$$\varepsilon = \frac{\Delta V}{G_f V_s} \quad (1)$$

$$\Delta V_g = V_g - V_{g \text{ tare}} \quad (2)$$

Where V_g is the measured unamplified bridge output voltage (V) and $V_{g \text{ tare}}$ is the unamplified bridge output voltage at the baseline condition (V) (2).

II. PROCEDURE

A. LabVIEW Virtual Instrument Setup

The SADI data acquisition device was controlled and voltage measurements taken throughout the experiment were processed using LabVIEW. The architecture included in the lab materials served as the basis for programming a virtual instrument. The program computed the values required to estimate the stool footrest's radius using measured voltages and provided geometric inputs [1].

The mean and standard deviation were computed using a set of voltage values obtained just before saving when a data point was recorded [4]. After being saved, these statistical results were then used in other calculations. The virtual instrument processed measured and tared voltages to calculate strain values, which were then used to estimate the footrest radius. Measurements based on the amplified voltage signal and the unamplified bridge voltage were calculated separately, and the resulting radius estimations were saved for further examination. The preceding 1000 voltage samples were used to calculate statistical values for each data point that was saved.

B. Data Acquisition System and WSB Integration

The measurement apparatus utilized in this experiment was constructed by connecting the SADI data gathering device, Wheatstone Bridge, and strain gauge. To record the source voltage, the DAQ's zero channel was linked to the proper bridge connections. One channel was connected to the amplifier output terminals to record the amplified voltage signal, while another channel was used to measure the unamplified bridge voltage. The LabVIEW virtual instrument received these voltage readings, analyzed them, and saved the results. Separate DAQ channels were used to capture the source voltage V_s , unamplified bridge voltage V_g , and amplified voltage V_{amp} , which were then supplied as inputs to the virtual instrument.

C. Sample Window Identification

To identify suitable voltage ranges for every test, several data collecting sample windows were examined. To find instances where signal saturation happened, preliminary measurements were made using a variety of sample window settings.

Saturation-producing sample windows were identified and excluded from data collection. These findings led to the selection of sample windows that avoided saturation while preserving enough resolution for precise voltage measurement. All the experiment's subsequent measurements were made during these chosen windows. Near-zero standard deviation in the recorded signal and voltage values that were getting close to the boundaries of the chosen input range were indicators of saturation.

D. Data Collection

The LabVIEW virtual instrument was used to gather voltage data after the data collecting system and sample windows were set up. With the feeler gauge still fastened around the base of the stool footrest, measurements were taken for the source voltage, the unamplified bridge voltage, and the amplified voltage signal. A collection of voltage values for every recorded data point was utilized to compute statistical quantities, which were then saved for further examination. These values were noted and saved for later examinations. With the feeler gauge fastened around the footrest of the stool, six measurement sets were taken. To create a baseline condition for taring, a second measure was taken with the gauge flat on a table. The physical configuration of the strain gauge and feeler gauge during data collection is shown in Fig. 2.



Fig. 2 Strain gauge mounted to a flexible feeler gauge and secured around the stool footrest during data collection.

E. Measurement of Circumference

A flexible measuring tape was used to take a direct measurement of the stool footrest's circumference. To measure the circumference, the tape was wound around the footrest. The results of the strain-based test were subsequently compared with this technique, which offered an independent way to assess the footrest radius. The divisions on the measuring tape were 0.1 cm, meaning that the reading uncertainty was ± 0.05 cm. Fig. 3 illustrates the location of the tape used to measure the circumference.



Fig 3. Flexible tape positioned around the stool footrest to measure circumference for the geometric radius estimate.

F. Data Processing and Calculations

$$\varepsilon_{V_{amp}} = \frac{\Delta V_{amp}}{G_f \text{ Gain } V_s} \quad (3)$$

Where ε_{amp} is the strain computed from the amplified bridge signal (mm/mm), ΔV_{amp} is the tared amplified output voltage (V), G_f is the strain gauge factor (unitless), Gain is the amplifier gain (unitless), and V_s is the bridge supply voltage (V).

$$r = \frac{d}{2\varepsilon} \quad (4)$$

Where r is the estimated outer radius of the stool footrest (mm), d is the offset distance between the strain gauge location and the neutral axis/center of curvature of the feeler gauge setup

(mm) (as defined in the lab handout), and ε is the measured strain (mm/mm).

$$r = \frac{C}{2\pi} \quad (5)$$

Where r is the estimated footrest radius (mm), C is the measured circumference of the footrest (mm), and π is estimated to 15 decimal places.

Using (1) and (2), mean strain values were calculated from tared voltage readings. Then, using (4), the corresponding radius estimates were determined. (5) was used to independently estimate the footrest radius based on the measured circumference. Mean values and standard deviations were calculated for every saved data point and saved for further examination.

III. RESULTS

Multiple data acquisition sample windows were used to gather voltage measurements to evaluate signal behavior and identify suitable voltage ranges for investigation. Using smaller sample windows resulted in saturation for some channel windows, especially for the amplified voltage signal. Saturation-producing sample windows were eliminated, and data collecting windows that avoided saturation and had adequate resolution were chosen. For every recorded data point, statistical variables such as the mean and standard deviation of each voltage signal were computed from 1000 samples. Table I provides a summary of the voltage measurements and associated statistical results for each sample window.

With the strain gauge still fastened around the base of the stool footrest, voltage readings for the source voltage, unamplified bridge voltage, and amplified bridge voltage were then taken using the chosen sample windows. The unamplified and amplified voltage signals showed detectable variability, although the source voltage stayed consistent throughout the trials. Each voltage measurement's standard deviation represented the trade-off between saturation avoidance and voltage resolution that was determined during sample window selection.

The unamplified bridge voltage data were utilized to compute strain values, which were then used to estimate the

TABLE I
MEAN VOLTAGES, STANDARD DEVIATIONS, & RESOLUTION FOR SELECTED SAMPLE WINDOWS

	<u>Channel 0</u>		<u>Channel 1</u>		<u>Channel 2</u>		
Sample Window [V]	Mean V_s [V]	Standard Deviation V_s	Mean V_g [V]	Standard Deviation V_g [V]	Mean V_{amp} [V]	Standard Deviation V_{amp} [V]	Resolution
± 0.64	0.639922	~ 0	0.000338	0.000059	0.015935	0.003933	0.000078125
± 1.28	1.279844	~ 0	0.000226	0.000079	0.020184	0.003985	0.00015625
± 2.56	2.49848	0.000142	0.001795	0.000021	0.020348	0.00391	0.0003125
± 5.12	2.497804	0.000087	-0.000001	0.000018	0.020152	0.003998	0.000625
± 10.24	2.496504	0.000247	-0.000098	0.000336	0.018168	0.004081	0.00125

stool footrest's radius. Table II reports estimated strain, intermediate voltage numbers, and the consequent radius estimates. There was more uncertainty since the unamplified voltage approach yielded radius estimates that varied noticeably between trials.

TABLE II
UNAMPLIFIED BRIDGE VOLTAGE & UNCERTAINTY

Parameter	Mean $\pm$ Uncertainty	Unit
V_g	0.0010 ± 0.00011	V
Std. V_g	0.000056	V
V_s	2.50 ± 0.00028	V
Std. V_s	0.00014	V
Delt. V_g	0.0011 ± 0.00015	V
ϵ_g	0.00086 ± 0.000088	mm/mm
rV_g	266.0 ± 37	mm

Measurements of amplified bridge voltage were also used to compute strain values. The amplified voltage approach demonstrated better sensitivity in comparison to the unamplified instance because of the larger signal magnitude. Using the amplified bridge voltage, the associated voltage measurements, strain values, and radius estimates are compiled in Table III. Comparing the amplified and unamplified methods, the former produced an estimated radius with less uncertainty. For physical interpretation, radius values are provided as magnitudes; negative strain denotes compressive bending.

TABLE III
AMPLIFIED BRIDGE VOLTAGE & UNCERTAINTY

Parameter	Mean $\pm$ Uncertainty	Unit
V_{amp}	-1.07 ± 0.0076	V
Std. V_{amp}	0.0039	V
V_s	2.5 ± 0.00028	V
Std. V_s	0.00014	V
ΔV_{amp}	-1.08 ± 0.011	V
ϵV_{amp}	-0.00092 ± 0.000012	mm/mm
rV_{amp}	250.6 ± 23	mm

A flexible measuring tape was used to measure the circumference directly, yielding an independent estimate of the stool footrest radius. A single geometric radius value was computed from the measured circumference and used as a benchmark to compare with strain-based techniques. Table IV provides a summary of the given values, tare voltages, and the circumference measurement used in the radius calculations.

The circumference-based measurement not only acts as a reference value but also provides insight into the shortcomings of strain-based radius estimating techniques. The circumference measurement is dependent on the physical positioning and

readability of the measuring tape, in contrast to the strain measurements, which are dependent on electrical signal conditioning, geometric parameters, and uncertainty propagation across numerous equations. Therefore, rather than electrical noise or component limits, the tape's resolution and operator reading dominate the uncertainty associated with this technology. Because of this difference, the circumference-based radius serves as a helpful standard for assessing the accuracy and sensitivity of the strain-derived measurements in the specified experimental setup.

TABLE IV
GIVEN VALUES & CIRCUMFERENCE MEASUREMENT

Parameter	Mean $\pm$ Uncertainty	Unit
t	0.4 ± 0.015	mm
d	0.03 ± 0.02	mm
G_f	2 ± 0.010	Unitless
Gain	900 ± 4.5	Unitless
$V_{g \text{ tare}}$	0.000083 ± 0.00010	V
$V_{amp \text{ tare}}$	0.0058 ± 0.0075	V
Flexible Tape	160.2 ± 0.05	cm

IV. DISCUSSION

In this experiment, the outside radius of a stool footrest was measured using three different methods, and the effects of data collection conditions on measurement quality were evaluated. A direct geometric measurement based on circumference, strain measurements from the amplified bridge voltage, and strain measurements from the unamplified bridge voltage were utilized to estimate the radius. Estimates of radius from the amplified strain method and circumference measurement were similar, however the unamplified strain approach displayed a greater degree of error. These results show how strain-based measurements are impacted by signal conditioning and data collection choices.

Signal resolution and uncertainty propagation are the primary reasons for discrepancies between the three radius estimates. Due to the small signal changes brought about by the unamplified bridge voltage, the resulting strain estimate was more vulnerable to ADC resolution limitations and noise. This sensitivity increased the uncertainty of the radius estimate derived from the unamplified voltage. However, by raising the effective signal magnitude, amplification reduced propagation uncertainty and enhanced strain resolution. When comparing the three methods using 95% confidence intervals, the amplified strain-based radius and the circumference-based radius agreed, while the unamplified strain-based radius displayed a broader confidence interval that partially overlapped with the other estimations. According to this comparison, amplification enhances the accuracy of strain-based radius measurements in the given circumstances.

Sample Window Selection

The choice of sample window had a significant effect on the voltage measurements' quality as well. Larger sample windows avoided saturation at the expense of lower resolution, whereas smaller windows improved voltage resolution but caused saturation in some channels. The chosen sample windows were a balance between these conflicting effects, guaranteeing that voltage signals stayed within the DAQ's detectable range while still having enough resolution to calculate strain. This trade-off emphasizes how crucial it is to assess data acquisition parameters before gathering experimental data.

Selection of Radius

Because of the higher signal magnitude that amplification provides, the amplified strain gauge method showed less uncertainty than the unamplified bridge voltage approach. Even with this improvement, the uncertainty of both strain-based radius estimations was higher than that of the radius determined by direct circumference measurement. Geometric characteristics and component tolerances, including the offset distance d , gauge thickness t , and gauge factor G_f , were the main sources of uncertainty in strain-based approaches. By propagating through the strain-radius relationship, these characteristics increased the impact of measurement variability. When the three approaches were compared using 95% confidence intervals, it was found that neither strain-based estimate completely overlapped with the geometric measurement. This demonstrated how sensitive digital strain measurements are to experimental setup and parameter uncertainty.

Limitations and Improvements

The precision of the strain-based radius estimates was impacted by several restrictions. Uncertainty was increased by assumptions of uniform curvature, inconsistent placement of the strain gauge, and inadequate contact between the feeler gauge and the footrest. Furthermore, the propagated uncertainty was dominated by component tolerances and geometric parameter uncertainty. Increasing the number of trials, fine-tuning gauge alignment and location, and utilizing more stable circuitry or higher-resolution data gathering gear could all help increase measurement accuracy.

V. CONCLUSION

In this study, digital data collecting techniques were employed to estimate the outer radius of a stool footrest using strain-based measures and direct geometric measurement. Radii obtained from both unamplified and amplified bridge voltage measurements have larger levels of uncertainty than those obtained using the flexible tape approach because of uncertainty propagation from geometric parameters, gauge factors, and voltage measurements. Amplification improved strain resolution and reduced uncertainty compared to the unamplified bridge voltage approach, highlighting the necessity of signal conditioning for the monitoring of small electrical signals. By finding a compromise between voltage resolution and saturation avoidance, the sample window selection directly affected measurement reliability. Overall, the comparison

of the three approaches confirmed the flexible tape measurement as the most accurate reference for this experiment and highlighted how sensitive strain-based measurements are to experimental setup and parameter tolerances.

APPENDIX

A. Analysis of Uncertainty

Uncertainty analysis was performed on all measured and calculated quantities associated with the three radius estimating methodologies. While Table V represents the uncertainties reported by the instrument and manufacturer for the given data, Table VI summarizes the uncertainties computed from several measurements. All calculated uncertainties were obtained using the Root Sum Squared (RSS) propagation approach, which accounts for the combined contribution of independent uncertainty sources.

B. Uncertainty in ΔV_g

The inaccuracies in the observed bridge voltage and its associated tare value are the source of the uncertainty in the tared bridge voltage, or ΔV_g . These contributions were combined using RSS uncertainty propagation as seen in (6)

$$\mu_{\Delta V_g} = \left[\left(\mu_{V_g} \left(\frac{\partial \Delta V_g}{\partial V_g} \right) \right)^2 + \left(\mu_{V_{gtare}} \left(\frac{\partial \Delta V_g}{\partial V_{gtare}} \right) \right)^2 \right]^{\frac{1}{2}} \quad (6)$$

C. Uncertainty of Strain from Unamplified Bridge Voltage

The tared bridge voltage, supply voltage, and strain gauge factor all affect the strain determined using the unamplified bridge voltage. By propagating the uncertainties of these parameters through the strain equation (7), the uncertainty related to this strain value was determined.

$$\mu_{\epsilon V_g} = \left[\left(\mu_{\Delta V_g} \left(\frac{\partial \epsilon}{\partial \Delta V_g} \right) \right)^2 + \left(\mu_{V_s} \left(\frac{\partial \epsilon}{\partial V_s} \right) \right)^2 + \left(\mu_{G_f} \left(\frac{\partial \epsilon}{\partial G_f} \right) \right)^2 \right]^{\frac{1}{2}} \quad (7)$$

D. Uncertainty in Radius from Unamplified Bridge Voltage

By propagating the uncertainties in the geometric parameters and the computed strain in (8), the uncertainty in the radius predicted using strain generated from the unamplified bridge voltage was found.

$$\mu_{rVg} = \left[\left(\mu_d \left(\frac{\partial r_{Vg}}{\partial d} \right) \right)^2 + \left(\mu_t \left(\frac{\partial r_{Vg}}{\partial t} \right) \right)^2 + \left(\mu_{\varepsilon Vg} \left(\frac{\partial r_{Vg}}{\partial \varepsilon_{Vg}} \right) \right)^2 \right]^{\frac{1}{2}} \quad (8)$$

E. Propagated Uncertainties for Unamplified Voltage

The propagated uncertainties associated with the unamplified bridge voltage method are summarized in Table VI.

F. Uncertainty of Tared Amplified Voltage

The uncertainty in the tared amplified bridge voltage, ΔV_{amp} , was determined from the uncertainties in the amplified voltage measurement and its tare value using RSS propagation in (9).

$$\mu_{\Delta V_{amp}} = \left[\left(\mu_{V_{amp}} \left(\frac{\partial \Delta V_{amp}}{\partial V_{amp}} \right) \right)^2 + \left(\mu_{V_{amptare}} \left(\frac{\partial \Delta V_{amp}}{\partial V_{amptare}} \right) \right)^2 \right]^{\frac{1}{2}} \quad (9)$$

G. Uncertainty of Strain from Amplified Voltage

The tared amplified voltage, amplifier gain, supply voltage, and strain gauge factor all affect the strain determined using the amplified bridge voltage. By using RSS propagation of these factors in (10), the uncertainty related to this strain value was determined.

$$\mu_{\varepsilon V_{amp}} = \left[\left(\mu_{\Delta V_{amp}} \left(\frac{\partial \varepsilon}{\partial \Delta V_{amp}} \right) \right)^2 + \left(\mu_{V_s} \left(\frac{\partial \varepsilon}{\partial V_s} \right) \right)^2 + \left(\mu_{G_f} \left(\frac{\partial \varepsilon}{\partial G_f} \right) \right)^2 + \left(\mu_{Gain} \left(\frac{\partial \varepsilon}{\partial Gain} \right) \right)^2 \right]^{\frac{1}{2}} \quad (10)$$

H. Uncertainty of Radius from Bridge Voltage

The uncertainty in the radius calculated using strain derived from the amplified bridge voltage was determined by propagating uncertainties in the geometric parameters and the amplified strain value in (11).

$$\mu_{rV_{amp}} = \left[\left(\mu_d \left(\frac{\partial r_{V_{amp}}}{\partial d} \right) \right)^2 + \left(\mu_t \left(\frac{\partial r_{V_{amp}}}{\partial t} \right) \right)^2 + \left(\mu_{\varepsilon V_{amp}} \left(\frac{\partial r_{V_{amp}}}{\partial \varepsilon_{V_{amp}}} \right) \right)^2 \right]^{\frac{1}{2}} \quad (11)$$

I. Propagated Uncertainties for Amplified Voltage

The propagated uncertainties associated with the amplified bridge voltage method are summarized in Table VI.

J. Uncertainty of Radius in Circumference Measurement

By propagating the uncertainty in the tape measurement, the uncertainty in the radius computed using the measured circumference was found. Because π is 15 decimal places, its uncertainty was deemed insignificant in comparison to the measurement uncertainty. (12) was used to calculate uncertainty.

$$\mu_{rc} = \left[\left(\mu_c \left(\frac{\partial r_c}{\partial c} \right) \right)^2 + \left(\mu_\pi \left(\frac{\partial r_c}{\partial \pi} \right) \right)^2 \right]^{\frac{1}{2}} \quad (12)$$

TABLE V
GIVEN UNCERTAINTIES

Parameter	Symbol	Uncertainty	Unit
Uncertainty of Gauge Factor	μ_{Gf}	± 0.010	Unitless
Uncertainty between Edge of Feeler Gauge and Center Strain Gauge	μ_d	± 0.02	mm
Uncertainty of feeler gauge thickness	μ_t	± 0.015	mm
Uncertainty of Gain	μ_{Gain}	± 4.5	Unitless
Uncertainty of Circumference	μ_c	± 0.5	mm

The primary sources of error related to manufacturer requirements, geometric measurements, and instrument resolution are represented by the uncertainties provided in Table V. The uncertainty propagation carried out in the following analysis uses these values as its main input. The contribution of each parameter to the final calculated uncertainties can be methodically assessed by first precisely identifying and measuring these distinct sources of error. This method guarantees that during propagation, more than one

source of uncertainty is neither unintentionally overlooked or counted twice.

By applying RSS propagation to the governing equations for voltage, strain, and radius, the uncertainties for all computed quantities were determined using baseline uncertainty values. Table VI provides a summary of the resulting propagated uncertainties, which represent the combined impacts of measurement variability and parameter limits.

TABLE VI
PROPAGATED UNCERTAINTIES

Parameter	Symbol	Uncertainty	Unit
Uncertainty of Delta bridge voltage	$\mu\Delta_{V_g}$	± 0.00015	V
Uncertainty of Delta bridge amplified voltage	$\mu\Delta_{V_{amp}}$	± 0.011	V
Uncertainty of strain using bridge voltage	$\mu_{\epsilon V_g}$	± 0.000088	mm/mm
Uncertainty of strain using amplified bridge voltage	$\mu_{\epsilon V_{amp}}$	± 0.000012	mm/mm
Uncertainty of Radius using bridge voltage	$\mu_{r V_g}$	± 37	mm
Uncertainty of Radius using amplified bridge voltage	$\mu_{r V_{amp}}$	± 23	mm
Uncertainty of radius using circumference	$\mu_{r_{circ}}$	± 0.080	mm

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